

## ROTATION INVARIANT BASES FOR LINEAR MODELLING OF OBSERVATIONS WITH A CIRCULAR ORDERING

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**Summary:** This paper is about the estimation of parameters of circularly ordered quantities, such as average monthly temperatures or the probability distribution of a discrete circular variable. We consider the case where the parameters are modelled as a linear combination of a subset of some orthonormal basis for  $\mathcal{R}^L$ , where  $L$  is the number of parameters. The subset may be chosen by means of a test of hypothesis or by some other selection procedure. We call the modelling procedure, that is the method used to select the subset and to estimate the parameters, rotation invariant if the model which is selected and fitted to the data is independent of rotations in the original data. We restrict our attention to linear parameterizations and show that rotation invariance can only be guaranteed if  $L$  is an integer power of 2. However, if the modelling procedure is modified by suitable pairing of the vectors in the basis, then rotation invariance can be attained for all integer  $L$ . This can be achieved using the well-known Fourier basis.

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### 1. INTRODUCTION

Many phenomena are quantified in terms of circular rather than linear variables. Examples of such variables from a wide variety of disciplines are given in Mardia (1972) and Batschelet (1981). In some applications, such as

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the study of animal orientation, wind direction or the orientation of fractures in rocks, the variable of interest is direction. In others it is time within a cycle of known period, such as the time of day when storms occur, when traffic densities peak, when the body temperature of animal is highest; the day of the year when birds begin to moult/nest/migrate; the month in which a disease is reported, and so on.

If such observations are grouped into, say,  $L$  intervals on a circle and the resulting frequencies are to be modelled by means of a probability distribution, then it seems reasonable to require that the modelling procedure be invariant with respect to rotations of the  $L$  categories. In other words the model which is selected and fitted should not depend on which particular category happens to be placed first if the categories are listed linearly.

A similar condition would apply to the estimation of other quantities which have a circular ordering but which may be linear or circular. Examples of these include average daily or monthly precipitation/temperature/evaporation, and seasonal factors for adjusting unemployment/inflation/sales indicators.

Four examples of application of the methods described in this paper are given in Bust (1987). These include models for hourly measurements of wind direction at Koeberg Nuclear Power Station, the orientation of flies of the species *Calliphara erythrocephala*, monthly counts of acute lymphatic leukaemia reported to British National Cancer Registration Scheme and weekly counts of storms at Botanic Gardens, Durban.

This paper gives the conditions under which rotation invariance is guaranteed for a certain class of modelling procedures for quantities that have a circular ordering. In particular we consider the case in which the  $L$  quantities (which may be probabilities or population parameters) are modelled as a linear combination of some subset of vectors of some orthonormal basis for  $\mathcal{R}^L$ , where the subset may be chosen by means of a test of hypothesis or by some other selection procedure. We assume that the basis contains the vector  $\phi_1 = (1/\sqrt{L})\mathbf{1}_L$  so that the remaining vectors in the

basis,  $\phi_2, \phi_3, \dots, \phi_L$ , form contrasts. We will call such (orthonormal) bases *contrast bases*.

The usefulness of such bases in statistical modelling is well-documented, see, for example, Kronmal and Tarter (1968), Ott and Kronmal (1976), Hall (1983), Diggle and Hall (1986), Liang and Krisnaiah (1985a), (1985b), Linhart and Zucchini (1986).

The vector  $\phi_1$  provides a constant term in the model, while the contrast vectors are used to introduce deviations from this constant. The modelling procedure is a systematic method of selecting which of the available contrasts should be included in the model to be fitted to the observations. It is assumed that  $\phi_1$  is always included.

This type of modelling procedure is mainly motivated by the necessity to reduce the number of parameters needed in the model. Suppose, for example, that we wished to estimate the average maximum temperature for each day of the year. This could be done by computing 365 averages over a number of years but, unless the data record is very large, these estimates would be relatively imprecise (have high standard errors). By reparameterizing the mean function in terms of a linear combination of some suitably constructed vectors, the number of parameters in the model can be substantially reduced, typically to less than 10 in examples of this type. By using a contrast basis the contribution of each contrast can be assessed separately for inclusion in the model. This leads to considerable computational savings.

In section 2 we investigate the conditions under which a contrast basis is rotation invariant. In particular we show that, for linear parameterizations, rotation invariance is guaranteed only if  $L$  is an integer power of 2. Now the values of  $L$  which are of interest in practice include 7 (for example the weekly cycle of power consumption), 12 (monthly data), 52 (weekly data), 73 (the number of 5-day groups in the year, or 'pentads', sometimes used in Hydrology), 365 (daily data). Since these integers are not powers of 2, this result is disappointingly restrictive. However, the situation can be remedied by restricting the modelling procedure a little. In section 3 we show that

rotation invariance can be guaranteed for all natural numbers  $L$  if the parameters of the model are suitably paired, and only the pairs are either included in or excluded from the model. We call this restricted type of invariance *pairwise rotation invariance*. The well-known Fourier basis is an orthonormal basis having this property.

## 2. ROTATION INVARIANCE

Our object in this section is to define rotation invariant (contrast) bases and then to characterize the class of such bases. We begin by introducing some notation.

We denote by  $R^n$  the operator which rotates the elements of a (column) vector  $n$  times from top to bottom. For example, if

$$(1) \quad \phi = \begin{bmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_{L-1} \end{bmatrix} = [\phi_i]_{i=0,1,2,\dots,L-1}$$

then  $R^n \phi = [\phi_{\{n+i\}}]_{i=0,1,2,\dots,L-1}$ , where the expression in curly brackets is taken modulo  $L$  (a convention which is used throughout the paper).

A contrast basis,  $(\phi_1, \phi_2, \dots, \phi_L)$ , for  $\mathcal{R}^L$  is an orthonormal basis with  $\phi_1 = (1/\sqrt{L})\mathbf{1}_L$ . Such a basis is called rotation invariant if for each vector  $\phi_q$  in the basis and for all rotations  $R^n$ ,  $n = 0, 1, \dots, L-1$ , there exists a (not necessarily distinct) vector  $\phi_r$  in the basis which is such that

$$R^n \phi_q \sim \phi_r,$$

where two vectors  $\phi$  and  $f$  are defined to be  $\sim$ -equivalent if  $\phi = \pm f$ .

The  $\sim$ -distinct rotations of a given contrast vector  $\phi$ , will be called the *rotation group* generated by  $\phi$  and denoted by  $G(\phi)$ , i.e.

$G(\phi) = \{R^n \phi : n = 0, 1, \dots, m-1 \text{ where } m \text{ is the smallest natural number satisfying } R^m \phi \sim \phi\}$ .

Rotation invariant model bases consist of the vector  $\phi_1 = (1/\sqrt{L})\mathbf{1}_L$ , and a number of rotation groups. The form of the basis is determined by that of the generators. Each generator must satisfy  $R^n \phi \sim \phi$  for some  $n \in \{1, 2, \dots, L-1\}$ . The nature of this restriction is indicated in the following result.

**Proposition 1.** Let  $\phi = [\phi_i]_{i=0, \dots, L-1}$ .

(a) If  $R^n \phi = +\phi$  for some  $n \in \{1, \dots, L-1\}$ , then  $\phi$  must be of the form

$$(2) \quad \mathbf{1}_d \otimes \begin{bmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_{(L/d)-1} \end{bmatrix}.$$

(b) If  $R^n \phi = -\phi$  for some  $n \in \{1, \dots, L-1\}$ , then  $\phi$  must be of the form

$$(3) \quad r_d \otimes \begin{bmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_{(L/d)-1} \end{bmatrix}$$

where in both cases  $d$  is the smallest natural number (a divisor of  $L$ ) satisfying  $dn \bmod L = 0$ , where in the second case  $d$  must be even and where

$$r_d = \mathbf{1}_{d/2} \otimes \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

(The symbol  $\otimes$  denotes the right Kronecker product.)

Part of the orthogonality restrictions on rotation groups is that all vectors within each rotation group must be orthogonal to one another. This

restriction cannot be met by contrast vectors of the form (2). (Proposition 2.) This means that only rotation groups generated by vectors of the form (3) need be considered. These have cardinality  $L/d$  where  $d$  is an even divisor of  $L$ . Thus rotation invariant contrast bases can only exist for even  $L$  (as odd  $L$  can have no even divisors). For the remainder of this section attention is restricted to even  $L$  and to generators which have the form given in (3).

**Proposition 3.** Let  $\phi$  be of the form (3) with  $L/d \geq 2$ . Then  $G(\phi)$  is orthogonal if

$$(R^n \phi)' \phi = 0 \text{ for } n = 1, 2, \dots, (L/d) - 1$$

which can be written in matrix form as

$$(4) \quad \begin{bmatrix} \phi_1 & \phi_2 & \cdots & \phi_{L/d-1} & -\phi_0 \\ \phi_2 & \phi_3 & \cdots & -\phi_0 & -\phi_1 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ \phi_{(L/d)-1} & -\phi_0 & \cdots & -\phi_{(L/d)-3} & -\phi_{(L/d)-2} \end{bmatrix} \begin{bmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_{(L/d)-1} \end{bmatrix} = \underline{0}.$$

If  $L/d = 2$  then  $G(\phi)$  is orthogonal. If  $L/d > 2$  and  $2d$  is a divisor of  $L$  then  $G(\phi)$  is orthogonal if

$$(5) \quad (R^n \phi)' \phi = 0 \text{ for } n = 1, 2, \dots, L/(2d) - 1$$

which is a consistent system.

**Proposition 4.** Two rotation groups of cardinalities  $L/d_1$  and  $L/d_2$  generated by vectors of the form (3) which are themselves orthogonal, are orthogonal to one another iff  $(d_1 + d_2)/gcd(d_1, d_2)$  is odd, with no further restrictions on the elements of the generators. (The function  $gcd(.,.)$  denotes the greatest common divisor of the two arguments.)

Proposition 4 makes the rather surprising statement that orthogonality between internally orthogonal rotation groups is a function of their cardinalities only. This condition is nevertheless restrictive. In particular it rules out the possibility of having two rotation groups of the same cardinality in a basis. The following proposition shows how restrictive the condition is.

**Proposition 5.** The maximum number of vectors which can be simultaneously included in a rotation invariant contrast basis of dimension  $L$  is less than or equal to  $L$ , with equality holding iff  $L = 2^m$  for some natural number  $m$ .

The theorem below which characterizes the form of rotation invariant contrast bases, follows from all of the above.

**Theorem 6.** Rotation invariant contrast bases of dimension  $L$  can exist only if  $L = 2^m$  for some natural number  $m$ , in which case the basis must be of the form

$$\left[ \frac{1}{\sqrt{2^m}} \mathbf{1}_{2^m}, G(r_2^m \otimes a), G\left[r_{2^{m-1}} \otimes \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}\right], \dots, G\left[r_2 \otimes \begin{bmatrix} p_1 \\ \vdots \\ p_{2^{m-1}} \end{bmatrix}\right] \right]$$

where each of the rotation group generators must be normalised and those of cardinality four or more must satisfy the orthogonality restrictions given in proposition 3.

### 3. PAIRWISE ROTATION INVARIANCE

Since the values of  $L$  which are of interest in practice include 7, 12, 52, 365 which are not integer powers of 2, it is necessary to modify the conditions under which theorem 6 was derived in order to accommodate such values of  $L$ . In this section we consider a modification to the notion of rotation invariance which preserves most of the desirable properties and which can be applied for an arbitrary number of categories.

To introduce the modification, consider grouping the vectors in some contrast basis into pairs, as follows:

$$(1/\sqrt{L} \mathbf{1}_L, (\phi_2, \phi_3), (\phi_4, \phi_5), \dots, (\phi_{L-1}, \phi_L))$$

for odd  $L$ , and

$$(1/\sqrt{L} \mathbf{1}_L, (\phi_2, \phi_3), (\phi_4, \phi_5), \dots, (\phi_{L-2}, \phi_{L-1}), \phi_L)$$

for even  $L$ . One has to be careful about the definition of these pairs of vectors. Two vector pairs,  $(\phi_1, \phi_2)$  and  $(f_1, f_2)$ , will be defined to be *non-distinct* if  $\phi_1 \sim f_1, \phi_2 \sim f_2$  or if  $\phi_1 \sim f_2, \phi_2 \sim f_1$ .

In the previous section the modelling procedures under consideration were those which provided fitted models comprising linear combinations of the vectors from some basis. Attention is now restricted to modelling procedures which provide models wherein each of the basis vector pairs is treated as a single unit, and the fitted model contains either both or neither of the vectors within a pair. In the case of even  $L$  the non-paired contrast vector is considered singly.

The structure of what follows is similar to that of the previous section. The starting point is a definition of pairwise rotation invariant bases and the end point is an explicit characterization of the form of all such contrast bases. It is convenient to begin by restricting attention to the simpler of the two cases, namely that of odd  $L$ . The theory for this case is then easily extended to incorporate even  $L$ .

The case where  $L$  is odd. For modelling procedures mentioned above rotation invariance can be guaranteed by using contrast bases that are "pairwise rotation invariant". The definition is a natural analog of that of a rotation invariant basis as given in the previous section.

A basis for  $\mathcal{K}^L$ , say  $(1/\sqrt{L} \mathbf{1}_L, (\phi_2, \phi_3), (\phi_4, \phi_5), \dots, (\phi_{L-1}, \phi_L))$ , is defined to be *pairwise rotation invariant* if for any pair in the basis

$$(\phi_{2q}, \phi_{2q+1}), q \in \{1, 2, \dots, (L-1)/2\}$$

and for all rotations  $R^n$  ( $n = 0, \dots, L-1$ ) there exists a pair

$$(f_{2r}, f_{2r+1}), r \in \{1, 2, \dots, (L-1)/2\}$$

in the basis, such that

$$(R^n \phi_{2q}, R^n \phi_{2q+1}) \approx (f_{2r}, f_{2r+1})$$

where  $\approx$ -equivalence is the pairwise analog of  $\sim$ -equivalence and is defined by  $(\phi_1, \phi_2) \approx (f_1, f_2)$  if

$$\phi_{i1} \phi_{j1} + \phi_{i2} \phi_{j2} = f_{i1} f_{j1} + f_{i2} f_{j2} \text{ for all } i \text{ and } j.$$

It is not difficult to show that two distinct vector pairs which are  $\approx$ -equivalent cannot appear in the same contrast basis (as all four vectors cannot each be orthogonal to one another). The only possibility then is that each vector pair must itself be responsible for its rotations, i.e. each vector pair must satisfy

$$(6) \quad (\phi_{2q}, \phi_{2q+1}) \approx (R^n \phi_{2q}, R^n \phi_{2q+1}) \text{ for } n = 0, \dots, L-1$$

which then becomes the effective definition of a pairwise rotation invariant basis.

The next objective is to investigate conditions under which pairwise rotation invariant model bases exist, and to characterize such bases. Note that as a special case of (6) one has that, for each  $q = 1, 2, \dots, (L-1)/2$  and for each  $i = 0, 1, \dots, L-1$

$$(\phi_{i,2q})^2 + (\phi_{i,2q+1})^2 = (\phi_{\{i+n\},2q})^2 + (\phi_{\{i+n\},2q+1})^2$$

for  $n = 0, \dots, L-1$ . This, together with the fact that  $\phi_{2q}$  and  $\phi_{2q+1}$  are normalised, implies that

$$(\phi_{i,2q})^2 + (\phi_{i,2q+1})^2 = 2/L \text{ for } i = 0, \dots, L-1; q = 1, \dots, (L-1)/2.$$

This suggests joining the two elements  $\phi_{i,2q}$  and  $\phi_{i,2q+1}$  into a single member of  $\mathcal{K}^2$ . Let

$$(7) \quad \underline{z}_q = [z_{iq}]_{i=0,\dots,L-1} = [\phi_{i,2q}, \phi_{i,2q+1}] \text{ for } q = 1, \dots, (L-1)/2.$$

Then for each  $q$  the elements of  $\underline{z}_q$  all lie on a circle of radius  $\sqrt{2/L}$ . It is more convenient to work with the polar representation of the  $z_{iq}$ , i.e.

$$(8) \quad z_{iq} = \sqrt{2/L} (\cos \arg z_{iq}, \sin \arg z_{iq})$$

where  $\arg z_{iq}$ , the *principal argument* of  $z_{iq}$ , is the angle which  $z_{iq}$  subtends at the origin with the horizontal axis, measured anti-clockwise and chosen such that  $0 \leq \arg z_{iq} < 2\pi$ . Since the length of each  $z_{iq}$  is known,  $z_{iq}$  is determined by the value of its principal argument.

Having established that for each  $q$ , the elements of  $\underline{z}_q$  all lie on the same circle, the full statement of (6) can be used to determine how they are spaced on the circle. Theorem 7 states that for each  $q$ , the  $z_{iq}$  ( $i = 0, \dots, L-1$ ) are equally spaced about the circle.

**Theorem 7.** A basis  $(1/\sqrt{L} \underline{1}_L, \underline{z}_1, \dots, \underline{z}_{(L-1)/2})$  where the  $\underline{z}_q$  are defined as in (8), is pairwise rotation invariant iff for  $q = 1, \dots, (L-1)/2$ ,

$$(9) \quad \arg z_{iq} = \{a_q i + b_q\} 2\pi/L \text{ for } i = 0, \dots, L-1$$

for some real numbers  $a_q$  and  $b_q$  with  $0 \leq a_q, b_q < L$ . The curly brackets (i.e. modulo  $L$ ) in (9) are redundant in the sense that

$$f(\{ai + b\} 2\pi/L) = f((ai + b) 2\pi/L)$$

where  $f$  is either  $\cos$  or  $\sin$ . Furthermore the  $b_q$  are redundant in the sense that different  $b_q$  do not produce  $\approx$ -distinct vector pairs. (Lemma 8).

Thus, at this stage, it is known that, up to  $\approx$ -equivalence, a pairwise rotation invariant basis, for odd  $L$ , can be taken to be of the form

$$(10) \quad (1/\sqrt{L} \mathbf{1}_L, [\sqrt{2/L} \cos(a_1 i 2\pi/L), \sqrt{2/L} \sin(a_1 i 2\pi/L)]_{i=0, \dots, L-1}, \dots, \\ [\sqrt{2/L} \cos(a_{\frac{1}{2}(L-1)} i 2\pi/L), \sqrt{2/L} \sin(a_{\frac{1}{2}(L-1)} i 2\pi/L)]_{i=0, \dots, L-1})$$

for some set of real numbers  $\{a_1, a_2, \dots, a_{\frac{1}{2}(L-1)}\}$  chosen such that the resulting basis is orthogonal. Two of the orthogonality conditions which the vectors in (10) must satisfy are:

$$\sum_{i=0}^{L-1} \cos(a_q i 2\pi/L) = 0 \quad \text{for all } a_q \quad (I)$$

$$\sum_{i=0}^{L-1} \sin(a_q i 2\pi/L) = 0 \quad \text{for all } a_q \quad (II)$$

**Lemma 9.** Let  $0 \leq a_q < L$ . Conditions (I) and (II) are satisfied simultaneously if and only if  $a_q \in \{1, \dots, L-1\}$ .

It follows that in order to construct an orthogonal pairwise rotation invariant basis we must in (10) choose  $\frac{1}{2}(L-1)$  unique  $a_q$  from  $\{1, \dots, L-1\}$ . Note that if  $a_q + a_r = L$  then

$$\cos(a_q i 2\pi/L) = \cos(a_r i 2\pi/L), \quad \sin(a_q i 2\pi/L) = -\sin(a_r i 2\pi/L)$$

so that the vector pairs

$$[\cos(a_q i 2\pi/L), \sin(a_q i 2\pi/L)]_{i=0, \dots, L-1}$$

and

$$[\cos(a_r i 2\pi/L), \sin(a_r i 2\pi/L)]_{i=0, \dots, L-1}$$

are non-distinct. Thus, without loss of generality, the  $a_q$  can be chosen to be the first  $(L-1)/2$  elements from  $\{1, \dots, L-1\}$  so that  $a_q = q$  for  $q=1, \dots, \frac{1}{2}(L-1)$ . It is well-known that this choice does lead to the basis in (10) being orthogonal. The theorem below follows directly.

**Theorem 10.** There is a pairwise rotation invariant model basis for each odd  $L$ , unique up to  $\approx$ -equivalence, namely

$$\begin{aligned}
 (11) \quad & (1/\sqrt{L} \, \underline{1}_L, \sqrt{2/L} [\cos(i2\pi/L), \sin(i2\pi/L)]_{i=0,\dots,L-1} \\
 & \sqrt{2/L} [\cos(2i2\pi/L), \sin(2i2\pi/L)]_{i=0,\dots,L-1}, \dots, \\
 & \sqrt{2/L} [\cos((L-1)/2 \, 2\pi/L), \sin((L-1)/2 \, 2\pi/L)]_{i=0,\dots,L-1}).
 \end{aligned}$$

**The case of even  $L$ .** We turn now to even  $L$  and deal with that non-paired last vector in the basis. Orthonormality and pairwise rotation invariance are obtained if for the paired vectors we use the sin-cos pairs as above, while the last vector is taken to be

$$[\cos((\frac{1}{2}L)i2\pi/L)]_i = \frac{1}{\sqrt{L}} [\cos(i\pi)]_i = \frac{1}{\sqrt{L}} r_L$$

This is in fact, the only vector, up to  $\sim$ -equivalence, with the required properties.

**Theorem 11.** Up to  $\approx$ -equivalence of the paired vectors and  $\sim$ -equivalence of the non-paired vector, there is a unique pairwise rotation invariant model basis for each even  $L$ , namely

$$\begin{aligned}
 (12) \quad & (1/\sqrt{L} \, \underline{1}_L, \sqrt{2/L} [\cos(i2\pi/L), \sin(i2\pi/L)]_{i=0,\dots,L-1}, \\
 & \sqrt{2/L} [\cos(2i2\pi/L), \sin(2i2\pi/L)]_{i=0,\dots,L-1}, \dots, \\
 & \sqrt{2/L} [\cos((L-1)/2 \, 2\pi/L), \sin((L-1)/2 \, 2\pi/L)]_{i=0,\dots,L-1}, (1/\sqrt{L})r_L).
 \end{aligned}$$

Bases of the form (11) and (12) are the well-known *Fourier* bases. We have established then

- (i) that a pairwise rotation invariant model basis exists for every dimension,
- (ii) that for each dimension there is essentially only one such basis; and
- (iii) for practical purposes this basis can be taken to be the Fourier basis.

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# APPENDIX

## A. Rotation invariance

### Proof of proposition 1

(a) Let  $\phi = [\phi_i]_{i=0, \dots, L-1}$  and let  $n$  satisfy  $1 \leq n \leq L-1$ . Then  $\phi = R^n \phi$  if and only if

$$(13) \quad \phi_i = \phi_{\{n+i\}} = \phi_{\{2n+i\}} = \dots = \phi_{\{(d-1)n+i\}} \text{ for } i = 0, 1, \dots, L-1$$

where  $d$  is as defined in the statement of the theorem, so that  $\phi_{\{dn+i\}} = \phi_i$  and all the indices in (13) are distinct. By the definition of  $d$ , there exists a natural number  $k$ ,  $1 \leq k < d$ , which is relatively prime to  $d$  such that  $n = kL/d$ . It can then be shown that

$$L/d = nk \bmod L.$$

Hence for  $i = 0, 1, \dots, L-1$

$$\begin{aligned} \phi_{\{i+L/d\}} &= \phi_{\{i+nk\}} \text{ with } 1 \leq k < d \\ &= \phi_i \end{aligned}$$

where the last equality follows from (13). Thus elements are repeated after every  $L/d$  and the result follows.

(b) Similarly,  $\phi = -R^n \phi$  iff

$$\begin{aligned} (14) \quad \phi_i &= (-1)\phi_{\{n+i\}} = (-1)^2\phi_{\{2n+i\}} = \dots = (-1)^{d-1}\phi_{\{(d-1)n+i\}} \\ &= (-1)^d\phi_{\{dn+i\}} \text{ for all } i. \end{aligned}$$

If  $d$  is odd (14) implies in particular that  $\phi_i = -\phi_i$  for all  $i$ , i.e.  $\phi = 0$  which cannot appear in a basis. Thus only even divisors  $d$  need be considered. By an argument similar to that used in part (a),

$$\phi_{\{i+L/d\}} = \phi_{\{i+nk\}} = (-1)^k \phi_i = -\phi_i \text{ for } i = 0, 1, \dots, L-1$$

where the last equality follows since  $k$  is odd, being relatively prime to the even number  $d$ . Hence, for even  $d$ , (14) holds if and only if

$$\phi_i = -\phi_{(L/d)+i} = \phi_{(2L/d)+i} = \dots = \phi_{((d-1)L/d)+i} \text{ for } i = 0, \dots, (L/d) - 1$$

and part (b) is proved.

### Proposition 2

Let  $\phi$  be a contrast vector which has the form given in (2) with  $L/d \geq 2$ . Then  $G(\phi)$  cannot appear in a rotation invariant contrast basis.

**Proof.** The rotation group  $G(\phi)$  is orthogonal if and only if

$$(15) \quad (R^n \phi)' \phi = 0 \text{ for } n = 1, 2, \dots, (L/d) - 1.$$

Since the elements of both  $\phi$  and  $R^n \phi$  are repeated after every  $L/d$  elements, (15) may be written as

$$\sum_{i=0}^{(L/d)-1} \phi_{\{n+i\}} \phi_i = 0 \text{ for } n = 1, 2, \dots, (L/d) - 1$$

or in matrix form

$$\begin{bmatrix} \phi_1 & \phi_2 & \cdots & \phi_{L/d-1} & \phi_0 \\ \phi_2 & \phi_3 & \cdots & \phi_0 & \phi_1 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ \phi_{(L/d)-1} & \phi_0 & \cdots & \phi_{(L/d)-3} & \phi_{(L/d)-2} \end{bmatrix} \begin{bmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_{(L/d)-1} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Pre-multiplying throughout by  $\frac{1}{(L/d)-1}$  gives

$$\begin{bmatrix} \sum_{j=0}^{(L/d)-1} \phi_j, & \sum_{j=0}^{(L/d)-1} \phi_j, & \dots, & \sum_{j=0}^{(L/d)-1} \phi_j, & \sum_{j=0}^{(L/d)-1} \phi_j \\ j \neq 0 & j \neq 1 & & j \neq (L/d)-2 & j \neq (L/d)-1 \end{bmatrix} \begin{bmatrix} \phi_0 \\ \vdots \\ \phi_{(L/d)-1} \end{bmatrix} = 0$$

which implies

$$(16) \quad \sum_{i=0}^{(L/d)-1} \sum_{\substack{j=0 \\ j \neq i}}^{(L/d)-1} \phi_i \phi_j = 0.$$

Given that  $\underline{\phi}$  is a normalised contrast vector, the expression on the left in (16) can be written as

$$\left[ \sum_{i=0}^{(L/d)-1} \phi_i \right]^2 - \sum_{i=0}^{(L/d)-1} \phi_i^2 = -1/d \quad \text{with } d \neq 0$$

which demonstrates the required contradiction.

### Proof of proposition 3

Let  $\underline{\phi}$  be of the form (3) with  $L/d \geq 2$ , i.e.  $\underline{\phi}$  has the form

$$\frac{1}{d/2} \otimes (\phi_0, \dots, \phi_{(L/d)-1}, -\phi_0, \dots, -\phi_{(L/d)-1})'$$

where the elements are repeated with alternate sign after every  $L/d$  elements.

Since  $R^{L/d} \underline{\phi} = -\underline{\phi}$  it follows that  $G(\underline{\phi})$  is orthogonal if

$$(17) \quad (R^n \underline{\phi})' \phi = 0 \quad \text{for } n = 1, 2, \dots, (L/d) - 1$$

which can be written as

$$(18) \quad \sum_{i=0}^{(L/d)-1} \phi_{\{n+i\}} \phi_i = 0 \quad \text{for } n = 1, \dots, (L/2d) - 1$$

where

$$\phi_{\{n+i\}} = \begin{cases} \phi_{(n+i)} & \text{for } n+i \leq (L/d) - 1 \\ -\phi_{(n+i) \bmod L/d} & \text{for } n+i \geq L/d. \end{cases}$$

Rewriting this in matrix form yields (4).

If  $L/d = 2$ , (4) reduces to

$$(\phi_1, -\phi_0) \begin{bmatrix} \phi_0 \\ \phi_1 \end{bmatrix} = 0$$

which is true.

In the case that  $L/d > 2$  and  $2d$  is a divisor of  $L$  it can be shown that some of the restrictions in (17) are redundant. In fact, for  $n = L/(2d)$ ,  $L/(2d) + 1, \dots, (L/d) - 1$

$$\begin{aligned} (19) \quad (R^n \phi)' \phi &= (R^{(L/d)-m} \phi)' \phi \quad \text{where } m = (L/d) - n, \\ &= \phi' R^{L-(L/d)+m} \phi = (R^{L/d} \phi)' (R^m \phi) = (-\phi)' (R^m \phi) \\ &= -(R^m \phi)' \phi \quad \text{where } 1 \leq m \leq L/(2d). \end{aligned}$$

Thus the restrictions in (17) for  $n \geq L/(2d)$  are equivalent to those for  $n \leq L/(2d)$ . Furthermore from the equality in (19) with  $n = L/(2d)$ , it follows that

$$(R^{L/(2d)} \phi)' \phi = 0$$

which implies that the restriction with  $n = L/(2d)$  is satisfied. Hence  $G(\phi)$  is orthogonal iff

$$(20) \quad (R^n \phi)' \phi = 0 \quad \text{for } n = 1, \dots, L/(2d) - 1.$$

The system is consistent since a non-zero solution can be obtained by putting  $\phi_0 = 1$  and  $\phi_2 = \dots = \phi_{(L/d)-1} = 0$ .

#### Proof of proposition 4

Let  $\phi$  and  $f$  be two rotation group generators of the form (3) whose cardinalities are  $L/d_1$  and  $L/d_2$  respectively and which satisfy the orthogonality condition given in proposition 3.

Let  $\underline{\alpha} = [\alpha_i]_{i=1, \dots, L}$  and  $\underline{\beta} = [\beta_i]_{i=1, \dots, L}$  be vectors from  $G(\phi)$  and  $G(f)$  respectively. The elements in  $\underline{\alpha}$  and  $\underline{\beta}$  are repeated with alternate sign after every  $L/d_1$  and  $L/d_2$  elements respectively. Let  $m$  denote the lowest common multiple of  $L/d_1$  and  $L/d_2$ . Then the elements of both  $\underline{\alpha}$  and  $\underline{\beta}$  are repeated (identically) after every  $2m$  elements and

$$\begin{aligned} \underline{\alpha}' \underline{\beta} &= L/(2m) \sum_{i=1}^{2m} \alpha_i \beta_i \\ &= L/(2m) \left[ \sum_{i=1}^m \alpha_i \beta_i + \sum_{i=1}^m \alpha_{(i+(md_1/L)L/d_1)} \beta_{(i+(md_2/L)L/d_2)} \right] \\ &= L/(2m) \left[ \sum_{i=1}^m \alpha_i \beta_i + \sum_{i=1}^m (-1)^{md_1/L} \alpha_i (-1)^{md_2/L} \beta_i \right] \\ &= L/(2m) (1 + (-1)^{(d_1+d_2)m/L}) \sum_{i=1}^m \alpha_i \beta_i. \end{aligned}$$

Using the fact that  $m = L/\gcd(d_1, d_2)$  it follows that  $G(\phi)$  and  $G(f)$  are orthogonal to each other if and only if  $(d_1 + d_2)/\gcd(d_1, d_2)$  is odd or

$$(21) \quad \sum_{i=1}^m \alpha_i \beta_i = 0 \quad \text{for all } \underline{\alpha} \in G(\phi), \underline{\beta} \in G(f).$$

It remains to show that condition (21) cannot be met by generators which generate orthogonal rotation groups.

For  $n = 0, \dots, ((L/d_1) - 1)$

$$(R^n \underline{\phi})' = r'_{d_1} \otimes (\phi_n, \dots, \phi_{(L/d_1)-1}, -\phi_0, \dots, -\phi_{n-1})$$

so that the first  $m$  elements of  $(R^n \underline{\phi})'$  are

$$r'_{m(d_1/L)} \otimes (\phi_n, \dots, \phi_{(L/d_1)-1}, -\phi_0, \dots, -\phi_{n-1}).$$

The first  $m$  elements of  $(R^n \underline{f})$  can be similarly expressed. Thus (21) can be written as

$$(22) \quad (r'_{md_1/L} \otimes A)' (r_{md_2/L} \otimes B) = 0_{L/d_1 \times L/d_2}$$

where

$$A = \begin{bmatrix} \phi_1 & \phi_1 & \cdots & \phi_{L/d_1-1} \\ \phi_1 & \phi_2 & \cdots & -\phi_0 \\ \vdots & \vdots & \cdots & \vdots \\ \phi_{(L/d_1)-1} & -\phi_0 & \cdots & -\phi_{(L/d_1)-2} \end{bmatrix} \quad B = \begin{bmatrix} f_0 & f_1 & \cdots & f_{(L/d_2)-1} \\ f_1 & f_2 & \cdots & -f_0 \\ \vdots & \vdots & \cdots & \vdots \\ f_{(L/d_2)-1} & -f_0 & \cdots & -f_{(L/d_2)-1} \end{bmatrix}.$$

Since  $G(\phi)$  is orthogonal it follows from proposition 3 that

$$A \begin{bmatrix} \phi_0 \\ \vdots \\ \phi_{(L/d_1)-1} \end{bmatrix} = \begin{bmatrix} 1/d_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

The analogous result holds for  $B$ . Thus pre-multiplying both sides of (22) with  $(\phi_0, \phi_1, \dots, \phi_{(L/d_1)-1})$  and post-multiplying with  $(f_0, f_1, \dots, f_{(L/d_2)-1})'$  gives

$$(23) \quad \left[ r_{md_1/L} \otimes \begin{bmatrix} 1/d_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right]' \left[ r_{md_2/L} \otimes \begin{bmatrix} 1/d_2 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right] = 0.$$

The expression on the left of this equality is equal to  $1/(d_1 d_2) > 0$  and the proof follows *reductio ad absurdum*.

### Proof of proposition 5

$L$  is taken to be even. Let the prime decomposition of  $L$  be

$$2^m P_1^{m_1} \dots P_k^{m_k}$$

where the  $P_i$  are distinct odd primes and  $m$  and the  $m_i$  are positive integers. Any even divisor of  $L$  is then of the form

$$2^n P_1^{n_1} \dots P_k^{n_k}$$

where  $1 \leq n \leq m$  and  $0 \leq n_i \leq m_i$  for  $i = 1, \dots, k$ , and gives rise to a rotation group of cardinality

$$(24) \quad \frac{L}{2^n P_1^{n_1} \dots P_k^{n_k}}$$

From proposition 4 it is known that two rotation groups of the same cardinality cannot appear in the same rotation invariant basis, and hence we can restrict attention to rotation groups of distinct cardinalities. For each fixed  $n$ , let  $C(n)$  denote the set of (distinct) cardinalities of the form (24) obtained by varying the  $n_i$  ( $i = 1, \dots, k$ ).

It is shown later that:

(25) for each  $n$  only one of the rotation groups with cardinalities in  $C(n)$  may be included in any model basis.

For each  $n$  the largest cardinality in  $C(n)$  is  $L/2^n$ . It happens that the set of rotation groups of cardinality  $L/2^n$  for  $n = 1, \dots, m$  can co-exist in the same basis. (To see this consider two internally orthogonal rotation groups of cardinality  $L/2^{n_1}$  and  $L/2^{n_2}$  with say  $n_1 > n_2$ .

Then

$$(2^{n_1} + 2^{n_2}) / \gcd(2^{n_1}, 2^{n_2}) = 2^{n_1 - n_2} + 1$$

which is odd, so that by proposition 4 the associated rotation groups will be orthogonal to each other.) This means that, including  $1/\sqrt{L} \underline{1}_L$ , the maximum number of vectors available for simultaneous inclusion in a rotation various basis, is

$$\begin{aligned} & 1 + L \sum_{n=1}^m 1/2^n \\ &= 1 + L(1 - 1/2^m) \\ &= L + (1 - P_1^{m_1} \dots P_k^{m_k}) \end{aligned}$$

and the proposition follows directly.

It remains to prove (25). Consider  $n$  fixed. Let  $L/d_1$  and  $L/d_2$  be two elements of  $C(n)$ , so that

$$\begin{aligned}d_1 &= 2^n P_1^{r_1} \dots P_k^{r_k} \\d_2 &= 2^n P_1^{s_1} \dots P_k^{s_k}\end{aligned}$$

for some  $r_i, s_i$ ;  $1 \leq r_i, s_i \leq m_i$  for  $i = 1, \dots, k$ . Then

$$\gcd(d_1, d_2) = 2^n P_1^{t_1} \dots P_k^{t_k} \text{ where } t_i = \min(r_i, s_i)$$

and

$$(26) \quad (d_1 + d_2)/\gcd(d_1, d_2) = (P_1^{r_1-t_1} \dots P_k^{r_k-t_k}) + (P_1^{s_1-t_1} \dots P_k^{s_k-t_k}).$$

In (26) each of the  $P_i$  is odd. Now any positive power of an odd number is odd and the product of two odd numbers is odd. Thus the two terms on the right-hand side of (26) are both odd; and their sum is even. Thus by proposition 4 the corresponding rotation groups cannot appear simultaneously in any model basis.

## B. Pairwise rotation invariance

### Proof of theorem 7

The basis is pairwise rotation invariant iff for  $q = 1, \dots, (L-1)/2$

$$(27) \quad [z_{iq}]_{i=0, \dots, L-1} \approx [z_{\{i+n\}, q}]_{i=0, \dots, L-1}$$

for  $n = 0, \dots, L-1$ . Expanding (27) according to the definition of  $\approx$ -equivalence and using the fact that for any two angles  $\theta_1$  and  $\theta_2$

$$(28) \quad \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 = \cos(\theta_1 - \theta_2)$$

yields

$$(29) \quad \cos(\arg z_{iq} - \arg z_{jq}) = \cos(\arg z_{\{i+n\},q} - \arg z_{\{j+n\},q}) \text{ for all } i \text{ and } j.$$

Fix  $q \in \{1, \dots, (L-1)/2\}$ . It is sufficient to show that (29) holding for  $n = 0, \dots, L-1$  is equivalent to

$$(30) \quad \arg z_{iq} = \{a_q i + b_q\}2\pi/L \text{ for some } a_q, b_q \in \mathbb{R} \text{ with } 0 \leq a_q, b_q < L$$

holding for  $i = 0, \dots, L-1$ .

Firstly, suppose that (30) holds for  $i = 0, \dots, L-1$ . Then for all  $i$  and  $j \in \{0, \dots, L-1\}$  and for any  $n \in \{0, \dots, L-1\}$

$$\begin{aligned} \cos(\arg z_{\{i+n\},q} - \arg z_{\{j+n\},q}) &= \cos[(\{i+n\}a_q + b_q - (\{j+n\}a_q + b_q))2\pi/L] \\ &= \cos(\arg z_{iq} - \arg z_{jq}) \end{aligned}$$

so that (29) holds.

For the reverse implication suppose that (29) holds for  $n = 0, \dots, L-1$  and prove that (30) holds for  $i = 0, \dots, L-1$  by induction on  $i$ . By hypothesis  $\cos(\arg z_{\{i+1\},q} - \arg z_{iq})$  is constant for all  $i$ . Hence for all  $i$  there exists a real number  $a_q$  with  $0 \leq a_q < L$  such that

$$(31) \quad \arg z_{\{i+1\},q} = \arg z_{iq} + a_q 2\pi/L + k_i 2\pi \text{ for some integer } k_i.$$

In addition note that  $b_q$  can be chosen such that  $\arg z_{0q} = b_q 2\pi/L$  with  $0 \leq b_q < L$ . With this choice of  $b_q$ , (30) is true for the case  $i = 0$ . Next suppose that (30) holds for some  $i \in \{0, \dots, L-2\}$ . Then using (31) it follows that

$$\begin{aligned} \arg z_{\{i+1\},q} &= (a_q i + b_q)2\pi/L + a_q 2\pi/L + k_i 2\pi \text{ for some integer } k_i \\ &= \{a_q (i+1) + b_q\}2\pi/L. \end{aligned}$$

Thus by induction, (30) holds for  $i = 0, \dots, L-1$ .

**Lemma 8.** For all  $a, b \in \mathbb{Z}$

$$\begin{aligned} [\cos((ai + b)2\pi/L), \sin((ai + b)2\pi/L)]_{i=0, \dots, L-1} \\ \approx [\cos(ai2\pi/L), \sin(ai2\pi/L)]_{i=0, \dots, L-1}. \end{aligned}$$

**Proof.** Put

$$\begin{aligned} T(a, b) = \{ \cos((ai + b)2\pi/L) \cdot \cos((aj + b)2\pi/L) \\ + \{ \sin((ai + b)2\pi/L) \cdot \sin((aj + b)2\pi/L) \}. \end{aligned}$$

It is sufficient to show that, for all  $a, b \in \mathbb{Z}$ ,  $T(a, b) = T(a, 0)$ . This follows immediately from (28).

**Proof of lemma 9.**

From the list of identities given in Bloomfield (1976, p. 15) one can establish that for  $0 \leq a < L$ :

$$\sum_{i=0}^{L-1} \cos(ai2\pi/L) = \begin{cases} \cos((L-1)a\pi/L) \sin(a\pi) / \sin(a\pi/L) & \text{if } a \neq 0, \\ L & \text{if } a = 0 \end{cases}$$

and

$$\sum_{i=0}^{L-1} \sin(ai2\pi/L) = \begin{cases} \sin((L-1)a\pi/L) \sin(a\pi) / \sin(a\pi/L) & \text{if } a \neq 0, \\ 0 & \text{if } a = 0. \end{cases}$$

It follows that (I) holds if and only if

$$a \in \{1, \dots, L-1\} \text{ or } a = L/(L-1) \cdot 1/2 \text{ or } a = L/(L-1) \cdot 3/2$$

while (II) holds if and only if

$$a \in \{0, \dots, L-1\} \text{ or } a = L/(L-1).$$

The result follows.

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